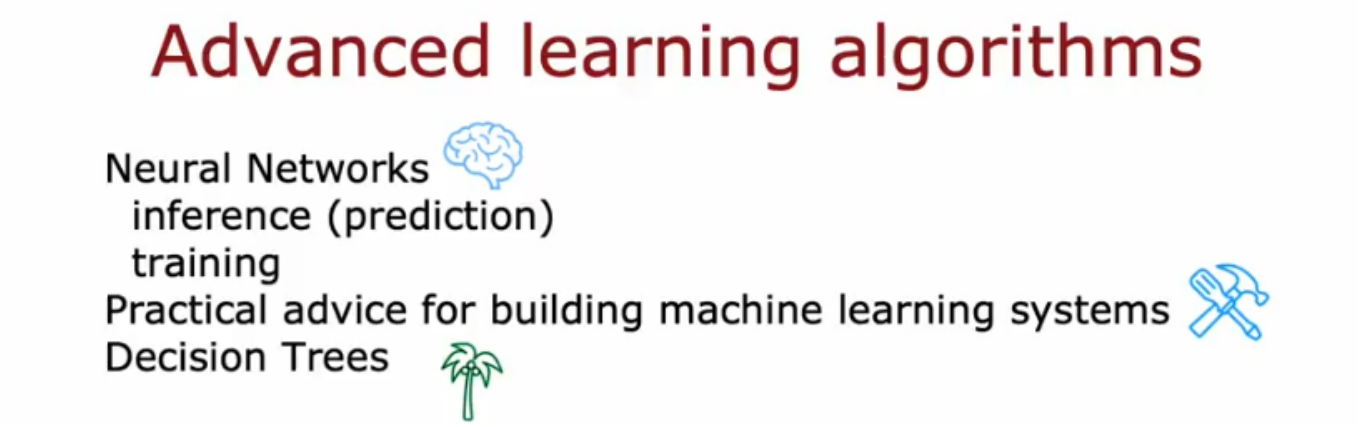
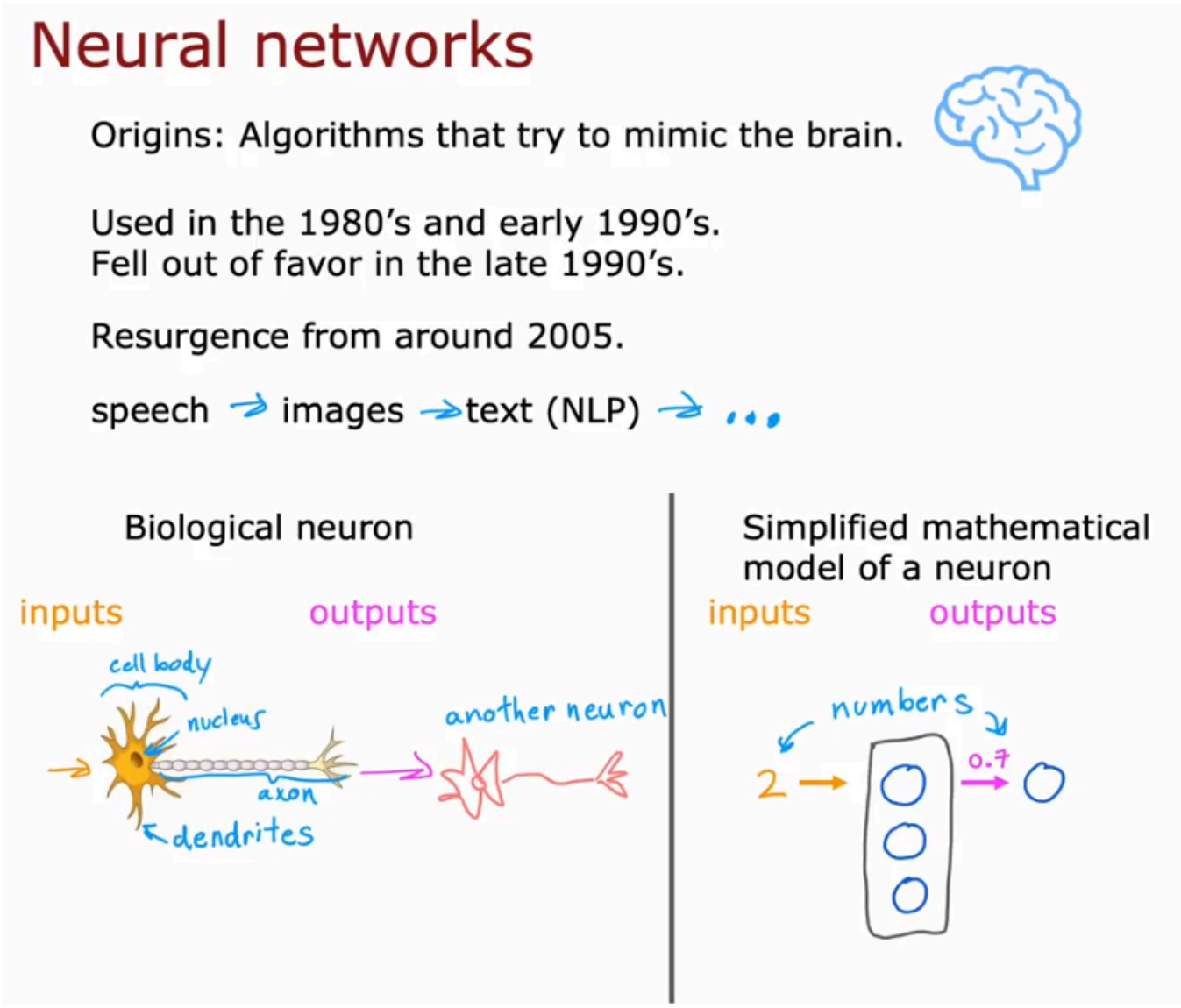
Day 12. Neural networks

> the first section in the specialization II: advanced learning algorithm

The 4 weeks course structure

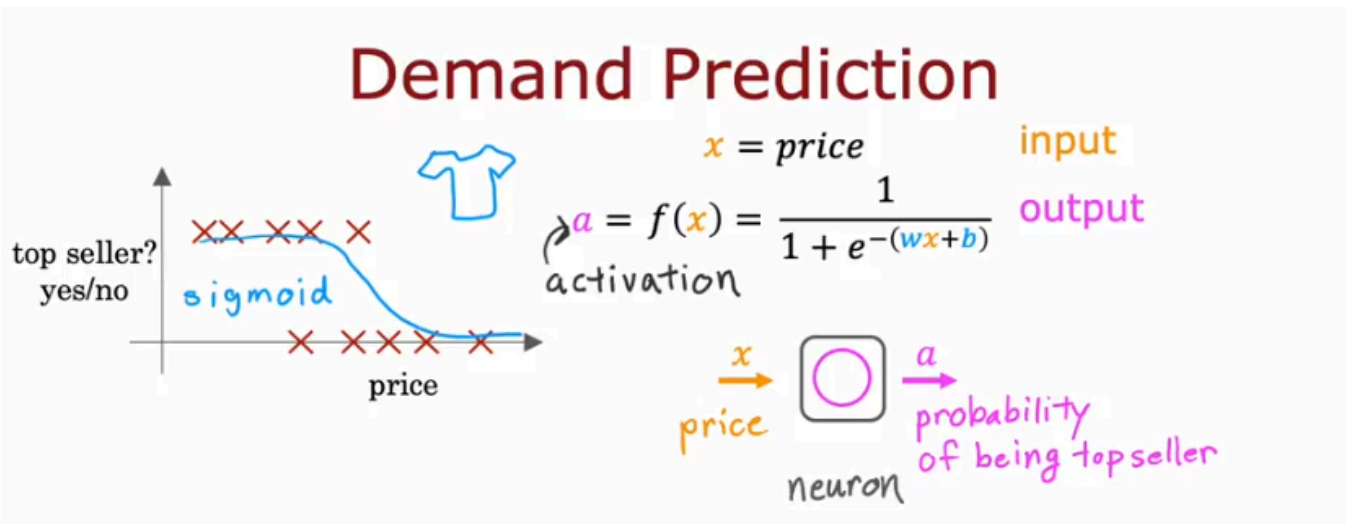


Neural netwoks intuition

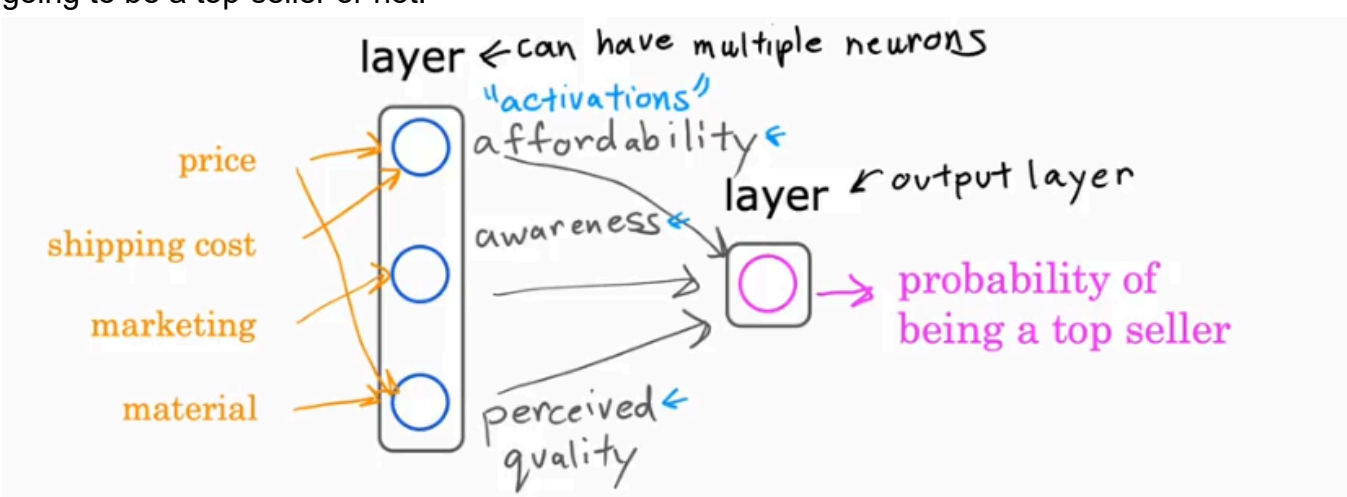


> even the origin is biological motivation, we will still discuss these problems in a engineering perspective.

Demand prediction



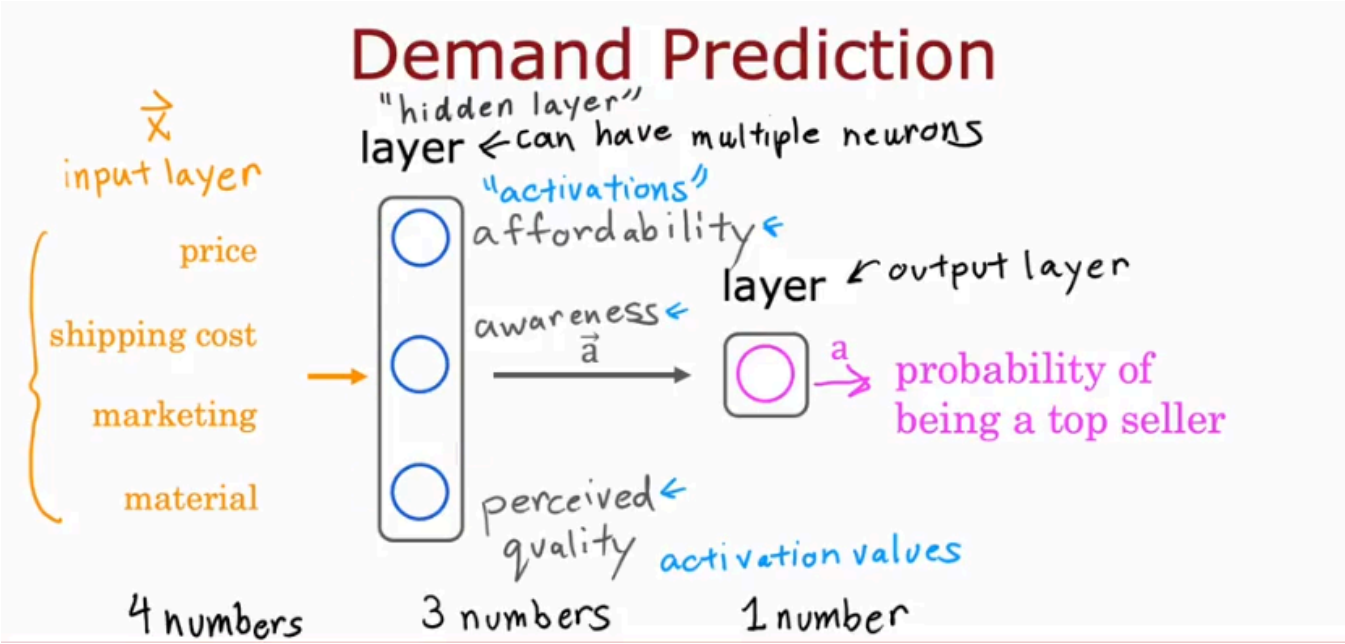
we only take one single input(feature) here to model the probablity of whether this T-shirt is going to be a top-seller or not.



it inputs four numbers, then the first layer computes the three activations, and then the output layer calculates the final output with the 3 activations.

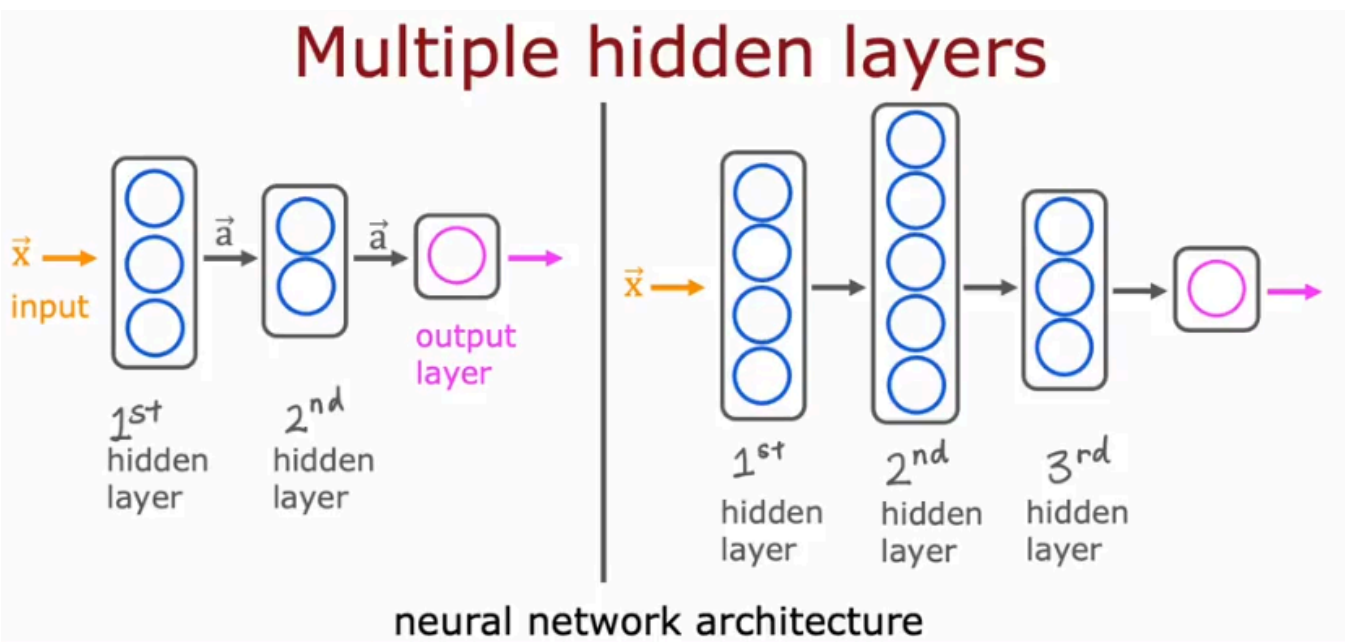
we also call the first 4 numbers as the input layer.

we can use vector to simplify the representation here.



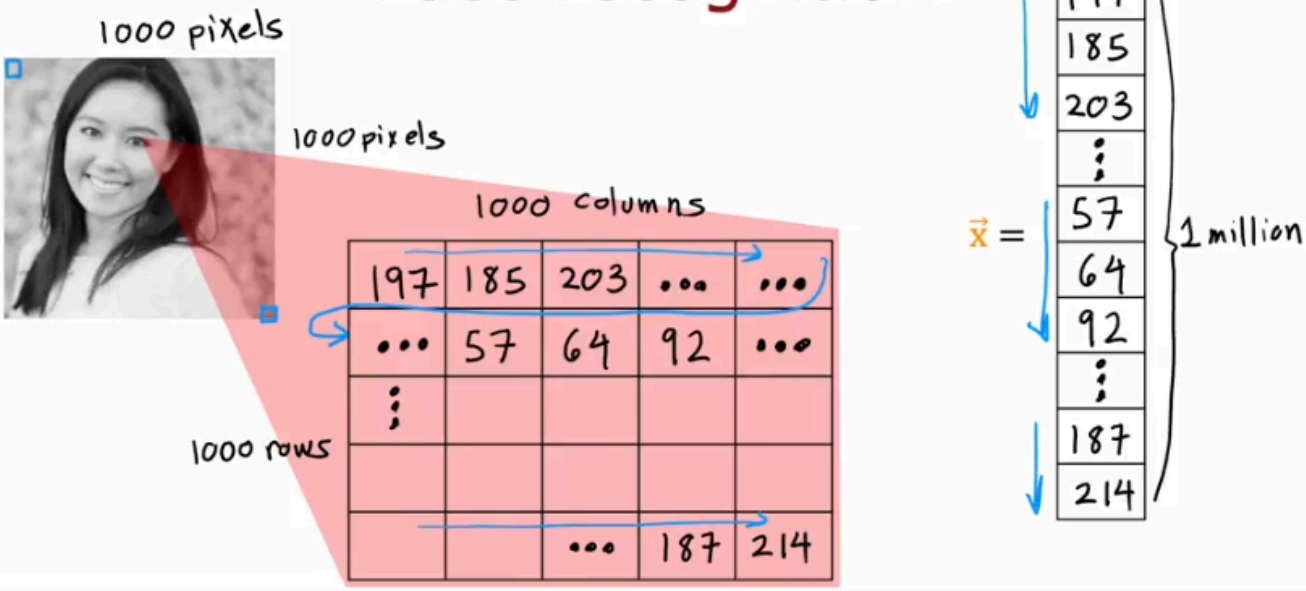
a little more terminology, we call the layer in the middle "hidden layers", we won't see them in the dataset.

one way to think of the neuro network is that it is actually using "better features" for new logistic regressions to get better results.

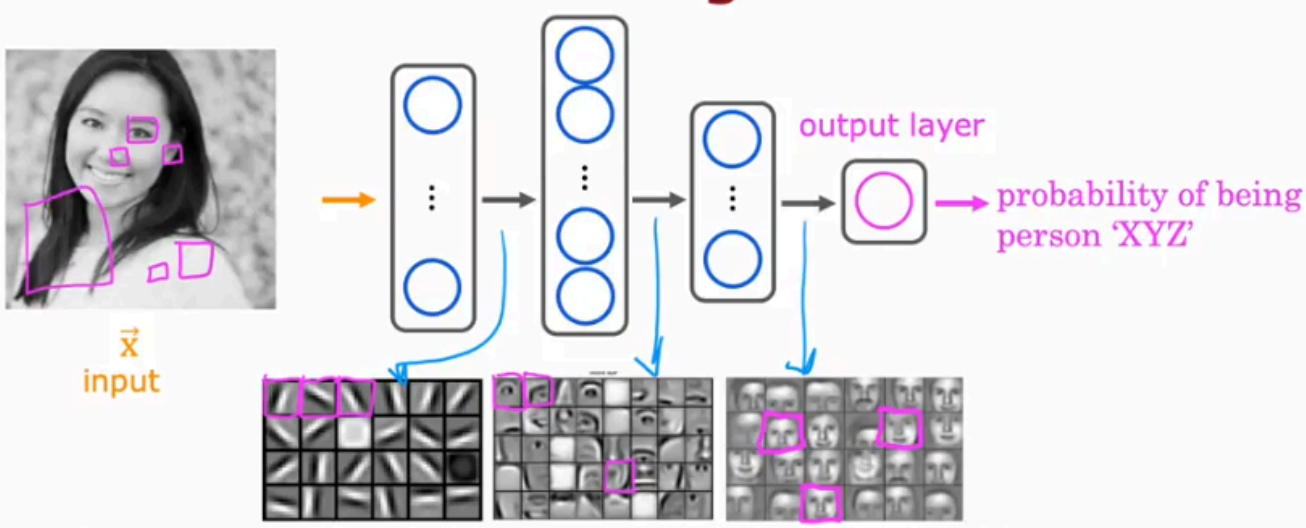


Example: recognizing images

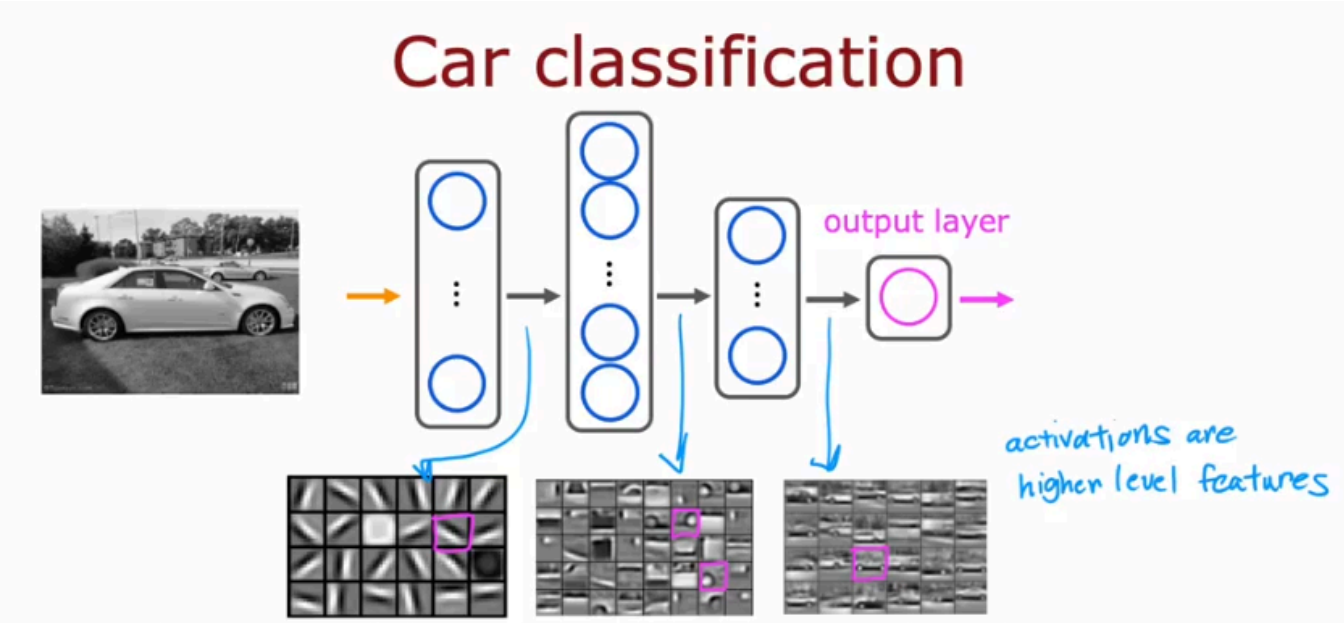
Face recognition



Face recognition



source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng



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Neural network layer

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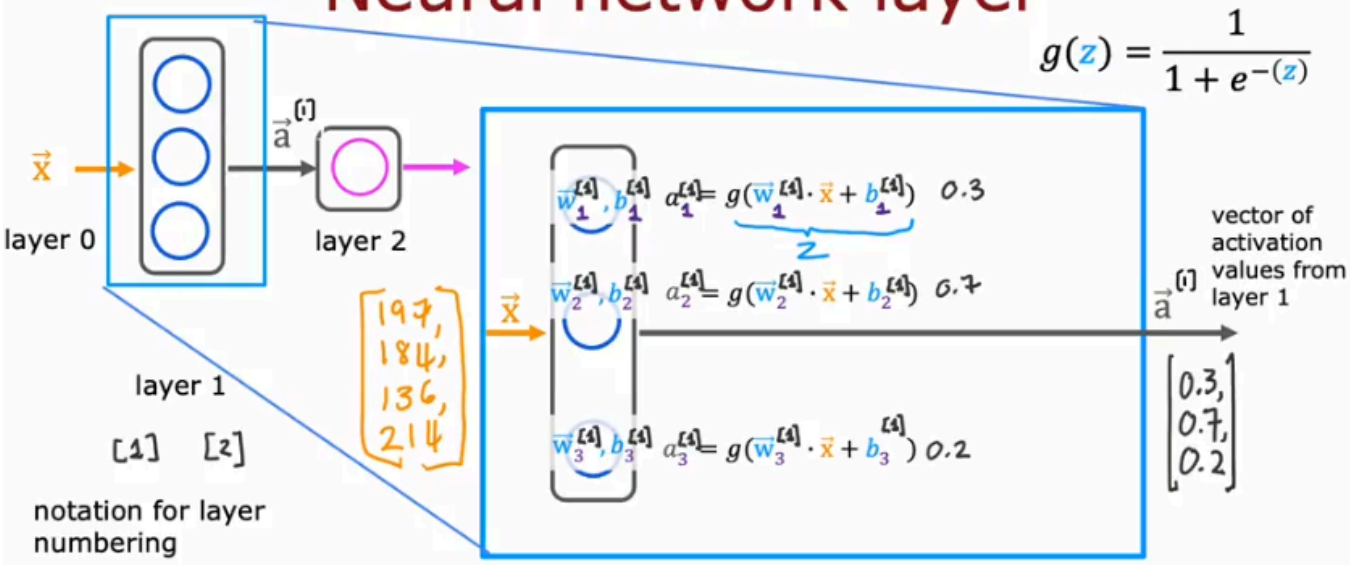
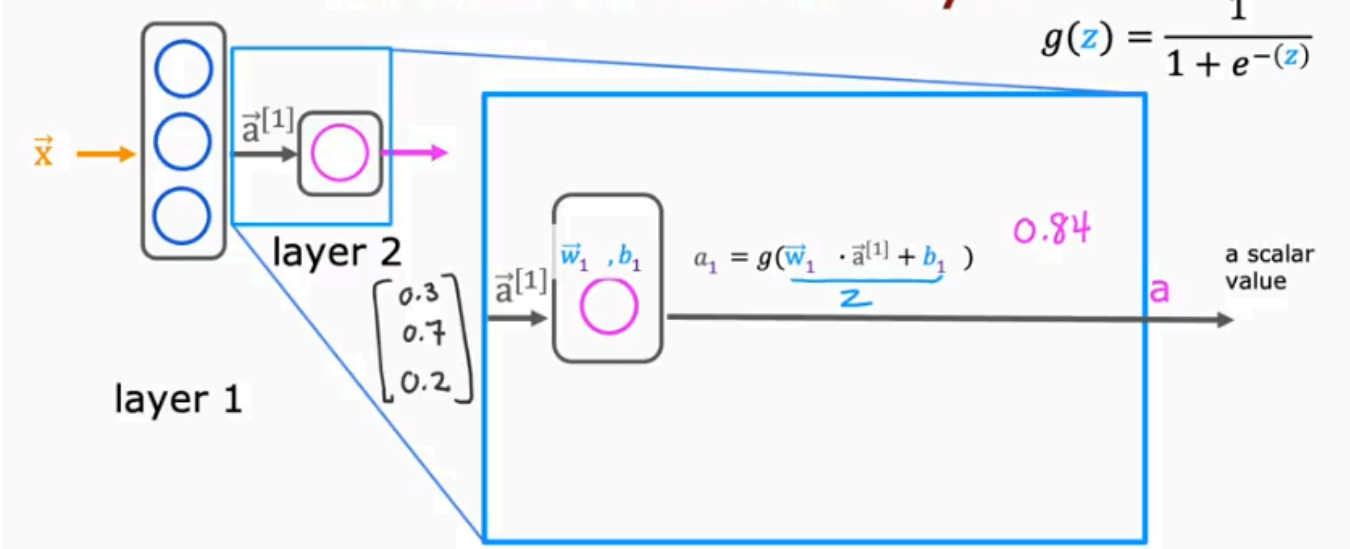
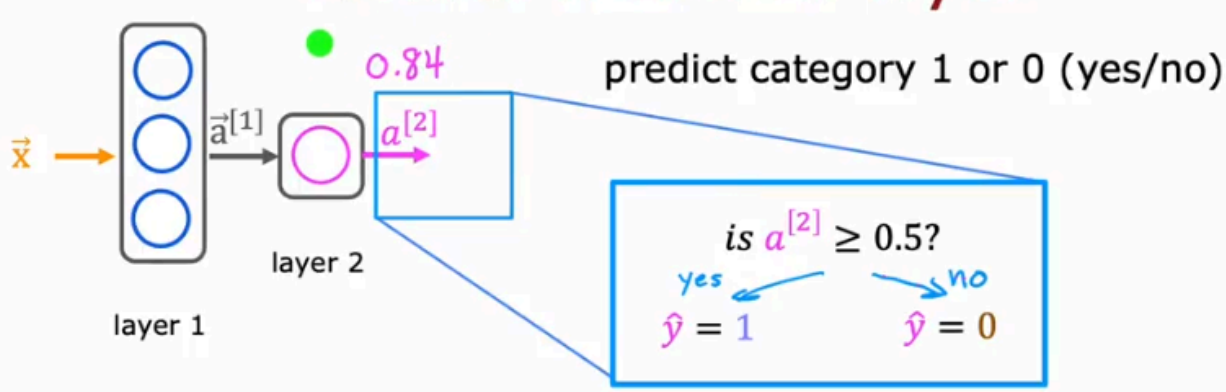


Neural network layer



Neural network layer